

Zhengyang Kris Weng

Robotic Systems R&D - Physical AI - Engineering

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Professional Summary

Robotics engineering tech lead with 5+ years developing surgical and domestic robotic systems at leading companies. Expert in robotic arm and actuator design, calibration systems, and control software with proven track record in system integration and R&D. M.S. in Robotics with focus on dexterous manipulation, robot control and machine learning. Strong background in mechanical design, embedded systems, and real-time control.

Professional Experience

Robotic Systems Engineer

Sunday Robotics, Mountain View, CA

Feb 2026 – Present

- **System Engineering:** Test, evaluate, and debug robotic components; develop motor and sensor drivers (C++/Python); design and develop robotic calibration and bring-up processes.
- **Scaling Policy Evaluation:** Triage and maintain autonomous fleet operation across ME/EE/FW/SW systems; support large-scale policy rollout and evaluation.

Senior Mechanical Engineer

Johnson & Johnson MedTech, Santa Clara, CA / Remote

Oct 2022 – Feb 2026

- **System Hardware R&D:** Core team member for v2 robotic arm development, contributing to system specifications, design reviews, and validation strategy for the Monarch robot surgery platform
- **Verification & Validation:** Developed test cases and fixtures and conducted V&V for surgical robotic devices, ensuring compliance with performance and safety requirements
- **Software Development:** Developed robot calibration software in C++ and designed admittance control visualization prototypes using Python and CoppeliaSim
- **Innovation & IP:** Invented fluid management system for Monarch Urology procedures (WO2025257720A1)
- **Leadership & Development:** Transitioned to remote part-time work while pursuing M.S. in Robotics (2024-2025), maintaining full project responsibilities and deliverables

Mechanical/Robotics Engineer

Neocis Inc., Miami, FL

Jun 2021 – Oct 2022

- **Mechatronics R&D:** Designed a 7-DoF robotic guidance arm for the Yomi S robot, including kinematic analysis and component specification. Developed a series of custom compact high-precision joint actuators from scratch
- **System Integration:** Led build and integration across three prototype arm iterations, achieving sub-millimeter accuracy for robotic dental surgery
- **Calibration Algorithms:** Developed optical robot kinematic calibration methods using optimization with regularization, enhancing accuracy and robustness
- **Leadership:** Mentored interns and delivered training to new hires on robotics hardware team
- **Career Progression:** Promoted to Senior Engineer (Aug 2022) in recognition of leadership and contribution

Mechanical Engineer Co-op

Harmonic Bionics Inc., Austin, TX

May 2020 – Dec 2020

- **Exoskeleton Development:** Designed robotic systems for 14-DoF rehabilitative upper extremity exoskeleton, including linear sizing mechatronic systems and handheld user interface
- **Operations:** Launched company machine shop, authored safety SOPs, and manufacturing equipment training

Education

M.S. in Robotics

Northwestern University, Evanston, IL Focus: Dexterous Manipulation, Kinematic Control, Imitation Learning

Sep 2024 – Aug 2025

B.S. in Mechanical Engineering

Georgia Institute of Technology, Atlanta, GA Concentration: Robotics and Control Systems

Sep 2016 – May 2021

Open-Source Projects

Hand Tracking Streamer: Meta Quest VR App for Tracking Hand Landmark Telemetry 2026
Zhengyang Kris Weng, San Jose, CA

- Developed a lightweight hand telemetry utility and companion SDKs that turn a Meta Quest headset into a precision controller for robotics teleoperation and motion capture. Available on Meta Horizon Store for free

LeVR: A Modular VR Teleoperation Framework for Imitation Learning in Dexterous Manipulation 2025
Northwestern University, Evanston, IL

- Developed modular VR teleoperation framework integrated with LeRobot platform for imitation learning in robotic manipulation
- Released open-source implementation (LeFranX) for Franka robotic arms and RobotEra XHand manipulator with high-frequency teleoperation and imitation learning
- Collected dataset of 100+ expert demonstrations and validated framework with state-of-the-art visuomotor policies

BiDexHand: Open-Source 16-DoF Dexterous Hand 2024-2025
Northwestern University, Evanston, IL

- Designed low-cost 3D-printed 16-DoF anthropomorphic hand with a complete ROS 2 control stack for joint-level control, functional retargeting, inverse kinematics, simulation and teleoperation functions
- Implemented vision-based calibration using AprilTags and integrated numerical IK solver with MoveIt!2 for real-time control with Franka robot arm

Technical Skills

Robotics & Simulation	Robot kinematics & dynamics, kinematic calibration, ROS 2, CoppeliaSim, Gazebo, MuJoCo, PyBullet, RViz, Unity
Software Development	Python (NumPy, Pandas, OpenCV), C++, MATLAB, Git, Linux, Bash
ML Frameworks	PyTorch, TensorFlow, HF Transformers, scikit-learn
Mechanical Design	SolidWorks (CSWE), OnShape, Fusion 360, ANSYS FEA, rapid prototyping, 3D printing (SLA, FDM, SLS, MJF)
Electrical	PCB design (KiCAD), oscilloscope, soldering
Machining	Milling, lathing, water jetting, laser cutting
Languages	English (native), Mandarin Chinese (native)

Publications & Patents

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- **Z.K. Weng**, M. Elwin, H. Liu, "LeVR: A Modular VR Teleoperation Framework for Imitation Learning in Dexterous Manipulation", *arXiv preprint*
 - **Z.K. Weng**, "BiDexHand: Design and Evaluation of an Open-Source 16-DoF Biomimetic Dexterous Hand", *IEEE ICRA 2025 Dexterity Workshop*, Spotlight Presentation
 - A. Harapanahalli*, E. Muly*, H. Welch*, T. Brumfiel*, **Z.K. Weng***, et al. "Towards a Biomimetic and Dexterous Robot Avatar: Design, Control, and Kinematics Considerations," *IEEE/ASME AIM 2020*, Best Breaking Results Poster (*equal contribution)
 - International Patent WO2025257720A1 — "Fluid Management System for Medical Procedures"
 - US Patent US11842304B2 — "Accessible Ride Hailing and Transit Platform"

Awards & Recognition

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- IEEE ICRA 2025 Dexterity Workshop — Spotlight Presentation for BiDexHand research
 - IEEE AIM 2020 — Best Late Breaking Results Poster for biomimetic robotics
 - First Place, 2021 Georgia Tech VIP Innovation Competition (Hardware, Devices & Robotics)
 - Georgia Tech President's Undergraduate Research Award (2021)
 - Toyota Mobility Foundation Challenge Winner (2017)
 - Certified SolidWorks Expert (CSWE) — Highest mechanical design certification awarded by Dassault Systèmes